Oxygenation, Fluid, Immune-Risk, and Cellular Evidence Assessment

NURSING · nursing-foundations · BoardPrep lesson v3

Official examination scope

PRC Resolution No. 10 s. 2025 and the August 2026 NLE program define examination scope, not topic weights. This lesson teaches a bounded oxygenation, fluid status, and immune-risk assessment foundation from the references below. Open RN and U.S. professional materials support general educational principles only; apply Philippine law, scope of practice, current clinical orders, referral pathways, and facility policy. This study material is not a clinical directive or proof of complete NP coverage.

Learning objectives

- Use a focused airway and breathing assessment to recognize deterioration, verify relevant
- observations, and escalate under current orders and local pathways.
- Integrate intake/output, perfusion, symptoms, and trends when assessing possible fluid
- imbalance without prescribing an unassessed fluid plan.
- Recognize that immunosuppression can change risk or presentation and that new symptoms warrant
- prompt assessment.
- Explain why a cellular diagnosis depends on an appropriately collected and processed specimen
- interpreted by a qualified pathologist with clinical context.
- Distinguish a visible image or screening clue from diagnostic evidence and communicate
- uncertainty without changing the canonical interpretation.

1. Read oxygenation as a changing clinical picture

Assess airway patency, breathing effort and pattern, ability to speak, skin and mental-status change, relevant measured trends, and the patient’s baseline. Verify a surprising reading and the sensor or measurement context when this can be done without delaying a safety response. New restlessness with worsening breathing or a downward oxygenation trend is a deterioration cue: position and support the patient within role, provide only currently prescribed oxygen, call for help under the facility pathway, and reassess response. Do not use a single number, device setting, or diagnosis as a substitute for the whole assessment. Numeric thresholds and U.S.-specific oxygen-order rules are intentionally omitted.

Sources for this section

Section reference: 8.2 Oxygenation Basic Concepts — Nursing Fundamentals 2e — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/chapter/8-2-basic-oxygenation-concepts/
Section reference: Basic Emergency Care: Approach to the Acutely Ill and Injured — World Health Organization and International Committee of the Red Cross — https://www.who.int/publications-detail-redirect/9789241513081

2. Combine symptoms and trends in fluid assessment

Fluid status is assessed from more than one cue: recent losses or intake, dizziness or weakness, perfusion and mental-status changes, urine-output trend, weight trend, examination, and ordered laboratory results. Ask about vomiting or other losses and compare findings over time; a single sign is not enough to identify cause or prescribe treatment. A cluster of vomiting, orthostatic dizziness, and markedly reduced output warrants prompt focused assessment and communication with the care team. Do not recommend unlimited fluids or independently change a prescribed restriction. Interpret measurements in context and follow local orders and escalation policy.

Sources for this section

Section reference: 15.2 Basic Fluid and Electrolyte Concepts — Nursing Fundamentals 2e — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/chapter/15-2-basic-fluid-and-electrolyte-concepts/
Section reference: Nursing Fundamentals — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/
Section reference: 15.6 Applying the Nursing Process — Nursing Fundamentals 2e — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/chapter/15-6-applying-the-nursing-process/

3. Treat new symptoms in immune-risk context

Risk and presentation depend on the disease and the specific treatment; the word “immunosuppression” alone does not identify a patient’s infection risk or an action threshold. The accessible National Cancer Institute guidance is limited to cancer and cancer treatment: it describes increased infection risk with some cancers and treatments such as chemotherapy, lists signs including fever and chills, and advises contacting the health care team. That page was reviewed in 2020, so its currentness requires review; this lesson does not reproduce its numeric threshold or import a U.S. pathway. Do not extend that source to every immune-modifying medicine. For a person receiving other immunosuppressive therapy, use current verified treatment-specific information and the authorized care plan. Assess a new change, communicate it through the current local pathway, and do not diagnose infection, change treatment, or wait for a self-selected threshold.

Sources for this section

Section reference: Infection and Neutropenia during Cancer Treatment — National Cancer Institute, U.S. National Institutes of Health — https://www.cancer.gov/about-cancer/treatment/side-effects/infection
Section reference: Basic Emergency Care: Approach to the Acutely Ill and Injured — World Health Organization and International Committee of the Red Cross — https://www.who.int/publications-detail-redirect/9789241513081

4. Connect cellular findings to specimen evidence

Cells are organized structures whose growth and function depend on regulated processes; cancer can arise when changes disrupt normal growth control. A photograph can show a surface or gross appearance, but it cannot establish what cells are present, how a specimen was processed, or what additional testing shows. The National Cancer Institute describes biopsy as removal of a tissue sample for a pathologist’s microscopic examination and possible additional testing. In practice, nurses help preserve specimen identity and chain-of-custody steps required by local policy, communicate what the patient has been told, and direct diagnostic questions to the responsible clinician. For example, a patient who sees an image of a lesion should be told that visual appearance alone is not a pathology diagnosis; the specimen and clinical assessment matter. This does not teach biopsy technique or predict an individual result.

Sources for this section

Section reference: What Is Cancer? — National Cancer Institute, U.S. National Institutes of Health — https://www.cancer.gov/about-cancer/understanding/what-is-cancer
Section reference: Tests and Procedures Used to Diagnose Cancer — National Cancer Institute, U.S. National Institutes of Health — https://www.cancer.gov/about-cancer/diagnosis-staging/diagnosis

Section reference: Surgical Pathology Reports — National Cancer Institute, U.S. National Institutes of Health — https://www.cancer.gov/about-cancer/diagnosis-staging/diagnosis/pathology-reports-fact-sheet

5. Read diagnostic evidence without overclaiming

A pathology report records findings from a defined specimen and is interpreted within the patient's broader clinical picture. The report may include specimen source, gross and microscopic descriptions, and a diagnosis; its wording is more than a photograph or a single visual impression. Screening tests, images, tissue examination, and molecular tests answer different questions and are not interchangeable. When a patient asks why a photograph cannot settle a biopsy result, explain that specimen processing and microscopy or other indicated tests provide evidence that a photograph alone cannot show, and that the clinician integrates those findings with symptoms and history. Report questions or a mismatch in the specimen label through the approved chain; do not revise a diagnosis or infer malignancy from appearance. NCI is a U.S. reference for general diagnostic concepts, not a Philippine diagnostic or treatment pathway.

Sources for this section

Section reference: Tests and Procedures Used to Diagnose Cancer — National Cancer Institute, U.S. National Institutes of Health — https://www.cancer.gov/about-cancer/diagnosis-staging/diagnosis
Section reference: Surgical Pathology Reports — National Cancer Institute, U.S. National Institutes of Health — https://www.cancer.gov/about-cancer/diagnosis-staging/diagnosis/pathology-reports-fact-sheet

Respond to worsening breathing

Question: A patient with pneumonia becomes restless, breathes faster, and has a falling oxygenation reading.
1. Assess airway and breathing, work of breathing, mental status, and the observed trend.
2. Verify measurement context if safe; position and provide only prescribed support within role.
3. Escalate through the facility pathway and reassess the response.
Answer: Treat the change as possible deterioration and seek timely help.
Check: No oxygen threshold, device setting, or independent order is supplied.

Assess fluid-loss cues

Question: A patient reports vomiting and dizziness and has a decreasing urine-output trend.
1. Ask about the timing and amount of losses and compare available intake/output and weight trends.
2. Assess perfusion, hydration cues, mental status, and relevant ordered results.
3. Report the pattern and response; follow the current care plan rather than prescribing unrestricted fluids.
Answer: Integrate the pattern and escalate concern; do not rely on one sign.
Check: Arithmetic of fluid totals does not independently establish cause or treatment.

Consider immune-risk context

Question: A person receiving immune-suppressing therapy reports a new fever.
1. Recognize increased risk and ask about onset, associated changes, and relevant treatment context.
2. Assess for deterioration and document objective findings.
3. Promptly notify the responsible team under the local pathway and reassess.
Answer: A new symptom in immune-risk context deserves prompt assessment and communication.

Check: Do not wait for a higher self-selected threshold or assume immune suppression prevents fever.

Explain why a photograph is not a biopsy result

Question: A client asks why a tissue sample cannot be interpreted from a phone photograph.
1. Separate visible appearance from evidence obtained from the collected specimen.
2. Explain that a pathologist examines processed tissue under a microscope and may use additional indicated tests.
3. Clarify that the responsible clinician relates the report to the patient's history and findings.
Answer: A photograph cannot show the processed tissue evidence and clinical interpretation needed for diagnosis.
Check: Do not infer benignity or malignancy from the picture or alter the answer key to fit it.

Handle a specimen-label discrepancy

Question: A label on a specimen container does not match the identifiers in the accompanying record.
1. Stop the local transfer/processing step that is within the nurse's control.
2. Follow the facility's specimen-identification and escalation policy; do not guess the correct identity.
3. Document the discrepancy and the verified handoff according to policy.
Answer: Preserve specimen identity by pausing and using the approved verification pathway.
Check: This example does not invent a laboratory procedure or local legal rule.

Common misconceptions

- A falling measurement is not interpreted without checking context, baseline, symptoms, and
- trend.
- A normal-looking single sign does not cancel worsening breathing or perfusion cues.
- Vomiting and low output do not justify unlimited fluids or an independent treatment change.
- Immunosuppression can alter risk or presentation; the absence of a high fever does not prove
- safety.
- A clinical mnemonic supports assessment but does not replace escalation or reassessment.
- A photograph, gross appearance, or screening clue alone is not interchangeable with
- microscopic specimen analysis and a pathology report.
- A pathology report concerns the submitted specimen; clinicians interpret it with the history
- and other findings rather than asking nursing learners to diagnose from an image.
- A specimen-label discrepancy is a verification and escalation problem, not permission to
- relabel by inference.

Practice patterns

- For respiratory change, assess the patient and trend, verify when safe, support within orders,
- escalate, and reassess.
- For fluid concerns, combine loss history, output, perfusion, mental status, and ordered data.
- For immune-risk cases, report new changes promptly without waiting for a numeric threshold.
- When a question asks why an image cannot establish a cellular diagnosis, identify the missing
- specimen processing and pathologist evidence.
- When specimen identity is uncertain, pause and use the authorized verification chain rather
- than guessing.

Official scope references

Official scope: Enhanced Tables of Specifications for the Subjects in the Nurses Licensure Examination — Professional Regulation Commission; Professional Regulatory Board of Nursing — https://www.prc.gov.ph/sites/default/files/2025-10%20%20published%20Enhanced%20TOS%20nuse.pdf
Official scope: Nursing Licensure Examination Program, August 2026 — Professional Regulation Commission; Professional Regulatory Board of Nursing — https://www.prc.gov.ph/sites/default/files/exam%20progam%20aug%202026%20nle%20(nursing)%20signed.pdf

Instructional references

Instructional reference: 8.2 Oxygenation Basic Concepts — Nursing Fundamentals 2e — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/chapter/8-2-basic-oxygenation-concepts/
Instructional reference: 15.2 Basic Fluid and Electrolyte Concepts — Nursing Fundamentals 2e — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/chapter/15-2-basic-fluid-and-electrolyte-concepts/
Instructional reference: 9.7 Applying the Nursing Process — Nursing Fundamentals 2e — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/chapter/9-7-applying-the-nursing-process/
Instructional reference: Basic Emergency Care: Approach to the Acutely Ill and Injured — World Health Organization and International Committee of the Red Cross — https://www.who.int/publications-detail-redirect/9789241513081
Instructional reference: Nursing Fundamentals — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/
Instructional reference: Core Infection Prevention and Control Practices — U.S. Centers for Disease Control and Prevention — https://www.cdc.gov/infection-control/hcp/core-practices/
Instructional reference: What Is Cancer? — National Cancer Institute, U.S. National Institutes of Health — https://www.cancer.gov/about-cancer/understanding/what-is-cancer
Instructional reference: Tests and Procedures Used to Diagnose Cancer — National Cancer Institute, U.S. National Institutes of Health — https://www.cancer.gov/about-cancer/diagnosis-staging/diagnosis
Instructional reference: Surgical Pathology Reports — National Cancer Institute, U.S. National Institutes of Health — https://www.cancer.gov/about-cancer/diagnosis-staging/diagnosis/pathology-reports-fact-sheet
Instructional reference: 15.6 Applying the Nursing Process — Nursing Fundamentals 2e — Open RN; Wisconsin Technical Colleges System — https://wtcs.pressbooks.pub/nursingfundamentals/chapter/15-6-applying-the-nursing-process/
Instructional reference: Infection and Neutropenia during Cancer Treatment — National Cancer Institute, U.S. National Institutes of Health — https://www.cancer.gov/about-cancer/treatment/side-effects/infection

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